

F112 — The NEW structural result from Casey's two threads: boundary $\leftrightarrow$ product, bulk $\leftrightarrow$ sum. The algebraic FORM of each lepton closed form is dictated by WHERE its norm integral lives. THREAD 1 (muon per-copy amplitude): the sixth power gets its correct home as a DETERMINANT over $\Lambda^2(T_p S^4)=\mathfrak{so}(4)$ (the six 2-forms at the measurement point) – $\det = \text{prod of 6 eigenvalues, each } (d_{\text{tau}}/d_{\text{muon}})/\text{vol}(S^4)=24/\pi^2$, giving $(24/\pi^2)^6$. The pi-counting FORCES the 6 (one π^2 per direction $\rightarrow \pi^{12}$) and $6 = \dim \mathfrak{so}(4) = C(4,2)$. HONEST: this FRAMES the per-copy amplitude as a determinant eigenvalue but does NOT derive that each eigenvalue $= 64/\text{vol}(S^4)$ – the full Szego (boundary) norm integral is OPEN. Found the right KIND of object (a product/determinant = what a boundary mode gives), not the value. THREAD 2 (tau vertex functional): $49*71 = g^{2(g+2)C_2} = g^3 + 2^{C_2}g^2 = 343+3136$ reads as a WEYL SPECTRAL COUNT $N(k) \sim V k^d + S k^{(d-1)}$ in $d=N_c=3$ dimensions (= characteristic multiplicity $a=n_C-2=N_c$ of D_{IV}^5), cutoff $k=g$, unit bulk + surface coefficient $2^{C_2}=64$. Exponents $\{3,2\}=\{d,d-1\}$ with $d=N_c=3$ is a real constraint the data satisfies. CAVEAT: lower Weyl terms (g^1, g^0) ABSENT $\rightarrow$ two leading terms only (large- g asymptotics or special cancellation, not explained); candidate mechanism with right form, not derived. THE TIE (new, defensible): muon = Shilov-BOUNDARY mode $\rightarrow$ boundary DETERMINANT $\rightarrow$ PRODUCT/power form $(24/\pi^2)^6$; tau = trivial rep $\rightarrow$ BULK spectral count over the whole domain $\rightarrow$ SUM form (volume+surface) $g^{3+2}C_2 g^2$. So WHY T190 is a power and T2003 is a sum is dictated by boundary-vs-bulk: boundary gives

products, bulk gives sums. Same soap-film/interior split, now in the algebra. Shared invariant: $2^C=64$ = muon's per-direction depth ($d_{\text{tau}}/d_{\text{muon}}$) AND tau's surface coefficient – one number, two roles. STRICT: closed NEITHER derivation; BANKS the boundary $\leftrightarrow$ product/bulk $\leftrightarrow$ sum complementarity (explains the algebraic shape of both forms) + 6 forced (π -count+dim so(4)); CANDIDATE = muon determinant eigenvalue $64/\text{vol}(S^4)$, tau Weyl-count ($d=N_c$, surface= 2^C); OPEN = muon Szego integral value, tau lower-terms/forcing; no re-fit; count 2.

Lyra (Claude Opus 4.8)

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F112 — Boundary gives products, bulk gives sums: the algebraic form of each lepton closed form

Casey: “Set up the per-copy amplitude, and take a run at the tau vertex functional.” I closed neither derivation — but the two threads together surfaced a new, defensible structural result about *why the closed forms have the shapes they do*.

Thread 1 — the muon per-copy amplitude: the right home, not the value

The sixth power gets its correct home: a determinant over $\Lambda^2(T_p S^4) = \mathfrak{so}(4)$ — the six 2-forms at the measurement point.

$$m_{\mu}/m_e = \det_{\{\Lambda^2(R^4)\}} M = \text{prod}_{\{6 \text{ two-forms}\}} (\text{eigenvalue}), \quad \text{eigenvalue} = (d_{\text{tau}}/d_{\text{muon}})/\text{vol}(S^4) = 24 \\ = (24/\pi^2)^6.$$

The π -counting still forces the six (one π^2 per direction $\rightarrow \pi^{12}$), and $6 = \dim \mathfrak{so}(4) = C(4,2) =$ the 2-planes in the tangent space at the measurement point.

Honest: this *frames* the per-copy amplitude as a determinant eigenvalue; it does not derive that each eigenvalue is $64/\text{vol}(S^4)$. That value is still asserted — the full Szegő (boundary) norm integral is the open piece. Thread 1 found the *right kind of object* — a product/determinant, which is exactly what a boundary mode gives — but not the eigenvalue from first principles.

Thread 2 — the tau vertex functional: a Weyl count, by its very shape

The tau is a sum, not a product, and that is the tell:

$$m_{\text{tau}}/m_e = g^2(g + 2^C) = g^3 + 2^C * g^2 = 343 + 3136.$$

This is exactly the shape of a Weyl spectral count $N(k) \sim V k^d + S k^{(d-1)} + \dots$ in dimension $d = N_c = 3$ (= the characteristic multiplicity $a = n_c - 2 = N_c$ of D_{IV}^5), - cutoff $k = g$, - unit bulk ($V g^3 = 343$) + surface coefficient $2^C C_2 = 64$ ($S g^2 = 3136$).

The exponents $\{3, 2\} = \{d, d-1\}$ with $d = N_c = 3$ are a genuine constraint the data satisfies — and $d = 3$ is the substrate's own characteristic multiplicity.

Caveat (not hidden): the lower Weyl terms (g^1, g^0) are absent — so this is the two *leading* terms only (large- g asymptotics, or a special cancellation), not explained. A candidate mechanism with the right form, not a derivation.

The tie — boundary $\leftrightarrow$ product, bulk $\leftrightarrow$ sum (the new, defensible result)

The two threads explain something neither closed-form's number did: why T190 is a power and T2003 is a sum.

The muon is a Shilov-boundary mode $\rightarrow$ its norm is a boundary determinant $\rightarrow$ a product / power form, $(24/\pi^2)^6$. The tau is the trivial rep, whose natural functional is a bulk spectral count over the whole domain $\rightarrow$ a sum form (volume + surface), $g^3 + 2^C C_2 g^2$.

So the algebraic form of each closed form is dictated by where the norm integral lives: a boundary integral is a product (a determinant over boundary directions); a bulk integral is a sum (a Weyl volume + surface count). This is the same soap-film / interior split Casey drew — now showing up in the algebra itself, not just the analysis. (The earlier reverse-engineering insight was *integer-vs-transcendental from the stratum*, F109/F110; this is its algebraic-form partner: *product-vs-sum from boundary-vs-bulk*.)

And the shared invariant is real and load-bearing in both: $> 2^C C_2 = 64$ is the muon's per-direction depth ($d_{\text{tau}}/d_{\text{muon}}$) AND the tau's surface coefficient — one number, two roles.

Strict tiering

- BANKS (defensible): the boundary $\leftrightarrow$ product / bulk $\leftrightarrow$ sum complementarity — the algebraic *shape* of both closed forms follows from boundary-vs-bulk; the muon's 6 is forced (π -count + $\dim \text{so}(4)$); $2^C C_2 = 64$ is the shared invariant (muon depth + tau surface coefficient).
- CANDIDATE (not banked): muon sixth power = determinant over $\text{so}(4)$ with eigenvalue $64/\text{vol}(S^4)$; tau = Weyl count in $d = N_c = 3$ with surface coeff $2^C C_2$.
- OPEN: the muon Szegő integral (the eigenvalue *value* from first principles); the tau Weyl forcing (unit bulk, the absent lower terms).
- NOT claimed / NOT fished: neither derivation is closed; no re-fit (T190/T2003 are inputs). Count stays 2.

Closure

Casey asked me to set up the muon per-copy amplitude and run at the tau vertex functional. I closed neither — but the attempt was worth more than a closure of one would have been, because it surfaced the rule behind both shapes. The muon, a Shilov-boundary mode, gives a product $((24/\pi^2)^6 = \text{a determinant over the six 2-forms } \text{so}(4) \text{ at the measurement point})$; the tau, the trivial rep, gives a sum $(g^3 + 2^C C_2 g^2 = \text{a Weyl volume+surface count in } d = N_c = 3)$. Boundary integrals are products; bulk integrals are sums — that is why one closed form is a power and the other a sum, and it is the soap-film/interior split written in the algebra. The shared $2^C C_2 = 64$ (muon depth = tau surface coefficient) ties them. What stays open is named: the muon eigenvalue value (the Szegő integral) and the tau Weyl forcing (the absent lower terms). Count stays honestly 2 — two candidate mechanisms with the right structural character, and one new defensible rule connecting them.

@Casey — did both; closed neither, but the pair gave a real rule. The muon (boundary) is a PRODUCT because a boundary norm is a determinant – the sixth power = $\det$ over the six 2-forms $\text{so}(4)$ at the point you measure. The tau (trivial rep) is a SUM because a bulk norm is a Weyl count – $49 \cdot 71 = g^3 + 2^C C_2 g^2 = \text{volume} + \text{surface in } d=N_c=3 \text{ dimensions, surface coefficient } 2^C C_2$. So WHY T190 is a power and T2003 a sum = boundary-vs-bulk: boundary gives products, bulk gives sums. Your soap-film/interior split, now in the algebra. Open: the muon eigenvalue (Szego integral) and the tau lower-terms. @Elie — your per-harmonic weight

= the muon determinant eigenvalue ($64/\text{vol}(S^4)$); and the tau as a Weyl count in $d=N_c=3$ to cutoff g is a toy you could test (does a 3d spectral count with surface coeff $2^{\wedge}C2$ reproduce $g^{3+2}C2 g^2$, and where do g^1, g^0 go?). @Grace — strict: closed NEITHER; BANKS boundary $\leftrightarrow$ product/bulk $\leftrightarrow$ sum + 6 forced + $2^{\wedge}C2$ shared; CANDIDATE det-eigenvalue & Weyl-count; OPEN Szego value & tau forcing; no re-fit; count 2. @Keeper — ledger: muon=boundary $\rightarrow$ det over $\text{so}(4)$ (6 two-forms) $\rightarrow$ PRODUCT $(24/\pi^2)^6$; tau=trivial rep $\rightarrow$ bulk Weyl count $d=N_c=3$ cutoff g unit-bulk+surface $2^{\wedge}C2\rightarrow$ SUM $g^{3+2}C2 g^2$; rule: boundary=products, bulk=sums; $2^{\wedge}C2=64$ shared (muon depth+tau surface); count 2.

— Lyra, Sat 2026-06-13 12:12 EDT (date-verified). F112 Casey's do-both. NEW: boundary $\leftrightarrow$ product, bulk $\leftrightarrow$ sum (algebraic form = where the norm integral lives). T1 muon: sixth power = det over $\text{Lambda}^2(T_p S^4)=\text{so}(4)$ (6 two-forms), eigenvalue $(d_{\text{tau}}/d_{\text{muon}})/\text{vol}(S^4)^{24/\pi^2}$; 6 forced $(\pi^{12}=(\pi^2)^6, =\dim \text{so}(4))$; HONEST frames per-copy as det eigenvalue, does NOT derive $64/\text{vol}(S^4)$ (Szego integral OPEN). T2 tau: $49*71=g^{2(g+2}C2)=g^{3+2}C2 g^2 =$ Weyl count $V k^d + S k^{(d-1)}$, $d=N_c=3$ (=char mult $a=n_c-2$), cutoff g , unit bulk+surface $2^{\wedge}C2=64$; exponents $\{3,2\}=\{d,d-1\}$; CAVEAT lower terms g^1, g^0 absent (asymptotics/cancellation, not explained). TIE: boundary=products (det), bulk=sums (Weyl); $2^{\wedge}C2=64$ shared. closed NEITHER; count 2.